Assignment Report

Part-1 Report

Summary

This comprehensive analysis of the M25 dataset reveals critical insights into workforce demographics, education patterns, work-life balance, and capital market participation. Through sophisticated data cleaning, feature engineering, and statistical analysis, we've uncovered actionable intelligence that can inform strategic decision-making in human resources, investment planning, and workforce development.

Key Findings at a Glance:

- **Work intensity shows stronger association with capital gains** ($\eta^2 = 0.0072$) than age groups ($\eta^2 = 0.0058$)
- **Significant age-work intensity relationship** ($p < 0.001$, $\chi^2 = 7108.05$)
- **95.3% data dimensionality reduction** achieved through intelligent feature grouping
- **Mid-career professionals dominate** high-intensity work categories

Data Transformation & Quality

Dataset Refinement Overview

Metric	Before	After	Improvement
Rows	48,843	48,842	99.998% retention
Columns	15	21	+40% feature enrichment
Missing Values	1 (Hours)	0	100% completion
Age Categories	74 unique	5 groups	**94.6% reduction**
Work Categories	96 unique	4 groups	**95.8% reduction**

Data Cleaning

1. **Age Validation & Correction**
    - Filtered out invalid ages (< 14 years)
    - Applied mode imputation for missing values
    - Maintained data integrity: 99.998% retention rate
2. **Work Hours Standardization**
    - Converted non-numeric entries to NaN
    - Applied statistical mode imputation

- Eliminated negative values through clipping

3. **Education Data Cleansing**

   - Trimmed whitespace inconsistencies
   - Removed rows with missing education data
   - Maintained comprehensive education coverage

Strategic Insights

1. Education Landscape Transformation

From Single to Grouped Categories

Education Level	Count	Percentage	Strategic Value
High School	24,060	49.2%	Core workforce foundation
Undergraduate	18,903	38.7%	Skilled professional base
Postgraduate	4,085	8.4%	Advanced expertise pool
School	1,711	3.5%	Entry-level potential
Kindergarten	83	0.2%	Minimal representation

Business Implication: Nearly 88% of workforce has high school or higher education - indicating strong foundational skills for advanced training programs.

2. Age-Career Stage Intelligence

Workforce Segmentation Strategy

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Age Distribution Across Career Stages:
├── Freshers (14-25): Young talent pipeline
├── Early Career (25-35): Skill development phase
├── Mid Career (35-55): Peak productivity zone
├── Late Career (55-65): Experience & mentorship
└── Senior (65+): Knowledge transfer opportunity
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3. Work Intensity Patterns: The Power Analysis

Critical Discovery: Work intensity drives capital engagement more than age

Work Category	Definition	Strategic Role
Full-Time	25-40 hrs/week	Core operational backbone
Overtime	41-60 hrs/week	High-value contributors
Part-Time	1-24 hrs/week	Flexible workforce
Excessive	61+ hrs/week	Burnout risk segment

Key Finding: Mid-career professionals (35-55) show highest engagement in overtime work (6,894 individuals), suggesting peak career commitment phase.

Capital Market Participation Analysis

Investment Behavior Insights

Capital Activity Distribution:

- **Active Participants:** Engage in capital gains/losses
- **Passive Population:** No capital market activity
- **Net Capital Impact:** Calculated as (Gains - Losses)

Statistical Significance Discovery

Work Intensity vs. Age Groups Impact on Capital Returns:

Factor	Eta-Squared (η^2)	Variance Explained	Ranking
Work Intensity	0.0072	0.72%	**Winner**
Age Groups	0.0058	0.58%	**Second**

Strategic Insight: Work intensity is 24% more predictive of capital performance than age group - focus investment programs on work behavior patterns rather than age demographics.

Work-Life Balance

Age-Work Intensity Correlation Matrix

Highly Significant Relationship ($p < 0.001$)

Full-Time Work Distribution:

- **Mid-Career leads:** 12,387 professionals (60% of category)
- **Early Career:** 8,034 professionals (strong engagement)
- **Freshers:** 5,207 professionals (building momentum)

Overtime Work Patterns:

- **Mid-Career dominance:** 6,894 professionals (peak performance)
- **Early Career intensity:** 3,469 professionals (career building)
- **Senior decline:** Only 235 professionals (lifestyle shift)

Feature Engineering Achievements

6 New Strategic Variables Created:

1. `Education_Level`: Grouped education for strategic analysis
2. `Age_Group`: Career stage segmentation
3. `Work_Intensity`: Work behavior classification
4. `Work_Life_Balance`: Lifestyle pattern identification
5. `Net_Capital`: Investment performance metric
6. `Any_Capital_Activity`: Market participation flag

Statistical Rigor

- **Chi-square test:** Confirmed age-work relationship ($\chi^2 = 7108.05$, $p < 0.001$)
- **Eta-squared analysis:** Quantified effect sizes for business decisions
- **Correlation analysis:** Validated relationships (Age-Hours: $r = 0.0716$)
- **Cross-tabulation:** Revealed detailed demographic patterns

Part-2 Report

Summary

This report provides a deep dive into the real estate market, leveraging the M25 Dataset 2 to uncover critical insights for investors. Through advanced data cleaning, feature engineering, and quantitative analysis, we have identified key trends in property valuation, location-based premiums, and the impact of developers and amenities. This analysis provides a strategic roadmap for navigating the complex real estate landscape.

Key Findings at a Glance:

- **Market Focus:** Mumbai dominates the high-end property segment, while Thane offers superior value for money (larger area for price).
- **Location Premium:** Prime locations command up to a 40% price premium, driven by price per square foot, not larger property sizes.
- **Value Drivers:** Luxury amenities like `Swimming Pool` and `Gymnasium` are significant price drivers in premium markets.
- **Developer Impact:** Top-tier developers are strongly associated with higher-priced properties and superior amenities, indicating brand value is a key factor.
- **Investment Timing:** Ready-to-move properties carry a premium for immediate availability, whereas under-construction properties offer a lower entry point with potential for higher appreciation.

Data Transformation & Quality

Dataset Refinement Overview

Our analysis is built on a foundation of meticulously cleaned and enriched data. Each analysis was conducted on a tailored data subset to ensure maximum relevance and accuracy.

Metric	Before	After (Typical)	Improvement
Rows	29,451	~20,000	~70% retention after targeted cleaning
Columns	66	+3	+4.5% feature enrichment
Missing Values	Widespread	0 in key columns	100% completion for analysis
Price Categories	Continuous	3 groups	Strategic Segmentation
Developer Tiers	1,000s unique	3 groups	Strategic Segmentation

Data Cleaning & Preparation

1. **Missing Value Imputation**:
    - Critical data points like `Price` and `Developer` with missing values were removed to maintain analytical integrity.
    - `Carpet Area` was imputed using the mean of its `Type of Property` group, preserving the distinct characteristics of different property types.
2. **Outlier Removal**:
    - **Z-Score Method**: Applied to the `Price` column to remove extreme, likely erroneous, data points that could skew the analysis.
    - **Regression-Based Removal**: Utilized to identify and remove properties where the price was an outlier relative to its carpet area.

Strategic Investment Insights

1. Price Segmentation & Market Overview

- **Approach**: We segmented properties into 'Low', 'Mid', and 'High' price ranges based on quantiles to analyze their distribution across cities and property types.
- **Findings**:
    - The high-end property market is heavily concentrated in Mumbai, with average prices significantly higher than in other regions.
    - Thane is a key player in the more diverse mid-range market, offering a good balance of price and amenities.
    - The low-range segment is widely distributed across various cities, indicating a broad base of affordable housing.
    - Investment strategies should be tailored to specific price tiers: Mumbai for premium investments, and Thane and other cities for mid-range opportunities.

2. City-Level Comparative Analysis (Mumbai vs Thane)

- **Approach**: A direct comparison of Mumbai and Thane across property types, carpet area, and price per square foot.
- **Findings**:
    - Mumbai properties are approximately 53.6% more expensive per square foot than those in Thane.
    - Thane offers larger properties on average (1024 sq ft) compared to Mumbai (871 sq ft).
    - **Strategic Choice for Investors**:
        - **Mumbai**: High-cost, high-appreciation potential, ideal for capital growth.
        - **Thane**: Better value with larger assets at a lower cost, suitable for rental yield and long-term value strategies.

3. Location-Based Premium Analysis

- **Approach**: We analyzed the price premium of properties flagged as being in a `isPrimeLocationProperty`.
- **Findings**:
    - Prime locations command a significant price premium, with properties in Mumbai being 40.5% more expensive per square foot and 27.0% more in Thane.
    - This premium is driven by the location ("address") itself, not by larger property sizes; prime properties are often smaller.
    - **Strategic Choice for Investors**:
        - **Prime Locations**: Buying into a premium brand with the promise of higher and faster capital appreciation.
        - **Non-Prime Locations**: Lower appreciation potential but may offer better rental yields due to lower acquisition costs.

4. Value-for-Money Opportunities

- **Approach**: We engineered a metric, `Area per Unit Price`, to identify properties offering the most space for the cost.
- **Findings**:
    - The best value for money is consistently found outside of Mumbai, with Thane leading the rankings.
    - For investors prioritizing asset size and value, emerging cities and the Thane region are the most attractive markets, offering lower entry points and high growth potential.
    - Residential houses and apartments generally offer the best value across different property types.

5. Feature & Amenity Impact on Price

- **Approach**: We analyzed the correlation between the presence of specific amenities and property prices in Mumbai and Thane.
- **Findings**:
    - Luxury amenities like `Swimming Pool`, `Gymnasium`, and `Club House` have a strong positive correlation with property prices, especially in the premium Mumbai market.
    - The presence of a swimming pool can increase a property's price by over 40%.
    - In high-end markets, luxury amenities are essential features that drive value, while in budget markets, the focus is on core amenities.

- Amenity strategies must be tailored to the specific target market segment.

6. Timeline & Readiness Effect on Pricing

- **Approach**: We compared the prices of `Ready to Move` properties versus those `Under Construction`.
- **Findings**:
  - Ready-to-move properties are, on average, 13.2% more expensive than those under construction, with the premium being more pronounced in Mumbai.
  - This premium is due to the convenience of immediate availability and the absence of construction-related risks.
  - **Strategic Choice for Investors**:
    - **Ready-to-Move**: A conservative, immediate-return strategy ideal for rental income.
    - **Under Construction**: A higher-risk, higher-reward growth strategy with a lower entry point and potential for greater capital appreciation.

7. Developer Impact on Properties

- **Approach**: We analyzed the relationship between property developers and prices, and categorized developers into tiers based on their average property prices.
- **Findings**:
  - A developer's brand is a strong indicator of a property's price and quality.
  - Premium developers consistently command higher prices and provide better amenities.
  - A developer's reputation for quality and luxury is a key factor in a property's market value.
  - Partnering with top-tier developers is a safer and more reliable investment strategy, as their brand equity contributes to the property's long-term value and market trust.

Feature Engineering Achievements

3 New Strategic Variables Created:

1. `Price Range`: For strategic market segmentation.
2. `Price per Sq Ft`: A standard industry metric for valuation.
3. `Area per Unit Price`: A custom metric to identify value-for-money deals.

Analytical Rigor

- **Descriptive Statistics**: Used to summarize the central tendency, dispersion, and shape of the dataset’s distribution.
- **Correlation Analysis**: To understand the relationship between variables like amenities and price.
- **Regression for Outlier Detection**: A sophisticated method to ensure data quality.
- **Data Visualization**: Extensive use of plots and charts to uncover and present trends and insights.